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NEGATIVE FEEDBACK THRESHOLD COMPENSATION FOR SENSE AMPLIFIERS

Abstract

Systems and methods are related to a memory device including a plate line. The memory device also includes a pair of ferroelectric layers implementing a pair of memory cells and coupled to opposite sides of the plate line. The memory device further includes a pair of digit lines each coupled to a respective ferroelectric layer of the pair of ferroelectric layers. The memory device also includes a sense amplifier coupled to the pair of digit lines and configured to sense and amplify voltages received at the pair of digit lines from the respective memory cells. The sense amplifier includes a pair of inverters configured to selectively couple to the pair of digit lines and a pair of switches each configured to cause a respective inverter of the pair of inverters to function as a unity gain amplifier during a compensation phase of the memory device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/559,626, filed Feb. 29, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND

Field of the Present Disclosure

[0002] Embodiments of the present disclosure relate generally to memory devices. More specifically, embodiments of the present disclosure relate to compensating for different characteristics of negative feedback amplifiers of sense amplifiers of a memory device.

Description of Related Art

[0003] Generally, a computing system may include electronic devices that, in operation, communicate information via electrical signals. For example, a computing system may include a processor communicatively coupled to a memory device, such as a dynamic random-access memory (DRAM) device, a ferroelectric random-access memory (FeRAM) device, another random-access memory (RAM) device, and/or a hybrid device that incorporates more than one type of RAM. In this manner, the processor may communicate with the memory device, for example, to retrieve executable instructions, retrieve data to be processed, by the processor, and/or store data output from the processor.

[0004] The memory devices utilize sense amplifiers during read operations. Specifically, the read circuitry of the memory device utilizes the sense amplifiers to receive low voltage (e.g., low differential) signals and amplify the small voltage (e.g., voltage differences) to enable the memory device to interpret the data properly. Some embodiments of the memory device may use multiple negative feedback/inverting amplifiers for a single cell. However, these different negative feedback/inverting amplifiers may have different characteristics, such as different thicknesses, aging, other general process variations, and/or any differences in the transistors used to form the negative feedback/inverting amplifiers even if the transistors are intended to be the same between the negative feedback/inverting amplifiers. These differences, even if relatively small, will be amplified during signal development and amplification. These amplified differences may negatively impact the operation of the sense amplifier potentially causing misinterpretation of data stored in the cells.

[0005] Embodiments of the present disclosure may be directed to one or more of the problems set forth above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a simplified block diagram illustrating certain features of a memory device having sense amplifiers and memory cells, according to an embodiment of the present disclosure;

[0007] FIG. 2 is a top view of lines of ferroelectric memory cells of FIG. 1 and a side view of the lines in the memory cells of FIG. 1, according to an embodiment of the present disclosure;

[0008] FIG. 3 is a circuit diagram of a connection of the lines of FIG. 2 to a sense amplifier of FIG. 3, according to an embodiment of the present disclosure;

[0009] FIG. 4 is a circuit diagram of the embodiment of the sense amplifier of FIG. 3 using a negative voltage and negative feedback amplifiers, according to an embodiment of the present

disclosure;

[0010] FIG. 5 is a graph and timing diagram of operation of the sense amplifier of FIG. 4, according to an embodiment of the present disclosure;

[0011] FIG. 6 is a diagram showing a representation of a negative feedback amplifier of FIG. 4 with a corresponding gain function, according to an embodiment of the present disclosure;

[0012] FIG. 7 is a circuit diagram of the embodiment of the sense amplifier of FIG. 3 using a ground voltage and negative feedback amplifiers, according to an embodiment of the present disclosure; and

[0013] FIG. 8 is a graph and timing diagram of operation of the sense amplifier of FIG. 7, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0014] One or more specific embodiments will be described below. To provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

[0015] Generally, a computing system may include electronic devices that, in operation, communicate information via electrical signals. For example, electronic devices in a computing system may include a processor communicatively coupled to memory. In this manner, the processor may communicate with memory to retrieve executable instructions, retrieve data to be processed by the processor, and/or store data output from the processor.

[0016] FeRAM memory technology utilization is increasing due to various possible market directions. As discussed below, FeRAM may be implemented with one-transistor, one-capacitor (1T1C) per cell or two-transistor, two-capacitor (2T2C) per cell arrangements. In the 2T2C arrangement, two cells are used for a single bit of data. One cell stores the data while another cell stores the complement of the data. The first cell is coupled to a digit line (DL) while the second is coupled to a digit line bar/false (DLB). A sense amplifier receives both signals from the DL and the DLB as a differential signal.

[0017] As discussed below, the 1T1C arrangement understandably uses less space than the 2T2C arrangement. However, the 2T2C accommodates faster speeds that may be used to obtain performances close or the same as DRAM specification requirements to enable the FeRAM to be compliant with DRAM specifications in dedicated FeRAM devices or hybrid DRAM/FeRAM devices. Thus, timing of various FeRAM operations may be important to comply with DRAM specified timings.

[0018] Furthermore, like DRAM, FeRAM utilizes a memory restore to restore memory values from latches back to the memory cells after a read. Specifically, each cell is programmed "0" and then cells that previously stored a "1" are programmed as such from the latches. Such programming of 1s may be conducted after writes are completed and may consume important time that is to be streamlined to comply with DRAM requirements.

[0019] Additionally, the differential signals from the DL and DLB to the sense amplifier may be relatively small (e.g., 50 mV-150 mV). Furthermore, due to the transistors used in the 2T2C or 1T1C arrangements, the operating voltage of the sense amplifier may be capped at a low level (e.g., 1.2 V). At such relatively small differences in voltages for the 2T2C arrangement, differences in the transistors (or the inverters they implement) may potentially cause issues in interpreting data from the memory cells. As discussed below, by enhancing and/or optimizing the response of the different inverters based on specific characteristics of different inverters differently may cause the fidelity of

the sense amplifiers to be improved.

[0020] Turning now to the figures, FIG. 1 is a simplified block diagram illustrating certain features of a memory device **10**. Specifically, the block diagram of FIG. 1 is a functional block diagram illustrating certain functionality of the memory device **10**. In accordance with one embodiment, the memory device **10** may be a double data rate type five synchronous dynamic random-access memory (DDR5 SDRAM) device, a ferroelectric memory (FeRAM) device, or a hybrid combination thereof. Various features of memory device **10** allow for reduced power consumption, more bandwidth and more storage capacity compared to prior generations of DDR SDRAM.

[0021] The memory device **10** may include a number of memory banks **12**. The memory banks **12** may be DDR5 SDRAM memory banks, FeRAM memory banks, or both, for instance. The memory banks **12** may be provided on one or more chips (e.g., SDRAM and FeRAM chips) that are arranged on dual inline memory modules (DIMMs). Each DIMM may include a number of memory chips (e.g., x8 or x16 memory chips), as will be appreciated. Each memory chip may include one or more memory banks **12**. The memory device **10** represents a portion of a single memory chip having a number of memory banks **12**. The memory banks **12** may be further arranged to form bank groups. For instance, for an 8 gigabyte (Gb) DDR5 SDRAM, the memory chip may include 16 memory banks **12**, arranged into 8 bank groups, each bank group including 2 memory banks. For a 16 Gb DDR5 SDRAM, the memory chip may include 32 memory banks **12**, arranged into 8 bank groups, each bank group including 4 memory banks, for instance. Various other configurations, organization, and sizes of the memory banks **12** on the memory device **10** may be utilized depending on the application and design of the overall system.

[0022] The memory banks **12** and/or bank control blocks **22** include sense amplifiers **13**. As previously noted, sense amplifiers **13** are used by the memory device **10** during read operations. Specifically, read circuitry of the memory device **10** utilizes the sense amplifiers **13** to receive low voltage (e.g., low differential) signals from the memory cells of the memory banks **12** and amplifies the small voltage differences to enable the memory device **10** to interpret the data properly.

[0023] The memory device **10** may include a command interface **14** and an input/output (I/O) interface **16**. The command interface **14** is configured to provide a number of signals (e.g., signals **15**) from an external (e.g., host) device (not shown), such as a processor or controller. The processor or controller may provide various signals **15** to the memory device **10** to facilitate the transmission and receipt of data to be written to or read from the memory device **10**.

[0024] As will be appreciated, the command interface **14** may include a number of circuits, such as a clock input circuit **18** and a command address input circuit **20**, for instance, to ensure proper handling of the signals **15**. The command interface **14** may receive one or more clock signals from an external device. Generally, double data rate (DDR) memory utilizes a differential pair of system clock signals, the true clock signal Clk_t and the bar clock signal Clk_c. The positive clock edge for DDR refers to the point where the rising true clock signal Clk_t crosses the falling bar clock signal Clk_c, while the negative clock edge indicates that transition of the falling true clock signal Clk_t and the rising of the bar clock signal Clk_c. Commands (e.g., read command, write command, etc.) are typically entered on the positive edges of the clock signal and data is transmitted or received on both the positive and negative clock edges.

[0025] The clock input circuit **18** receives the true clock signal Clk_t and the bar clock signal Clk_c and generates an internal clock signal CLK. The internal clock signal CLK is supplied to an internal clock generator, such as a delay locked loop (DLL) circuit **30**. The DLL circuit **30** generates a phase controlled internal clock signal LCLK based on the received internal clock signal CLK. The phase controlled internal clock signal LCLK is supplied to the I/O interface **16**, for instance and is used as a timing signal for determining an output timing of read data. In some embodiments, the clock input circuit **18** may include circuitry that splits the clock signal into multiple (e.g., 4) phases. The clock input circuit **18** may also include phase detection circuitry to

detect which phase receives a first pulse when sets of pulses occur too frequently to enable the clock input circuit **18** to reset between sets of pulses.

[0026] The internal clock signal(s)/phases CLK may also be provided to various other components within the memory device **10** and may be used to generate various additional internal clock signals. For instance, the internal clock signal CLK may be provided to a command decoder **32**. The command decoder **32** may receive command signals from the command bus **34** and may decode the command signals to provide various internal commands. For instance, the command decoder **32** may provide command signals to the DLL circuit **30** over the bus **36** to coordinate generation of the phase controlled internal clock signal LCLK. The phase controlled internal clock signal LCLK may be used to clock data through the IO interface **16**, for instance.

[0027] Further, the command decoder **32** may decode commands, such as read commands, write commands, mode-register set commands, activate commands, etc., and provide access to a particular memory bank **12** corresponding to the command, via the bus path **40**. As will be appreciated, the memory device **10** may include various other decoders, such as row decoders and column decoders, to facilitate access to the memory banks **12**. In one embodiment, each memory bank **12** includes the bank control block **22** which provides the necessary decoding (e.g., row decoder and column decoder), as well as other features, such as timing control and data control, to facilitate the execution of commands to and from the memory banks **12**.

[0028] The memory device **10** executes operations, such as read commands and write commands, based on the command/address signals received from an external device, such as a processor. In one embodiment, the command/address bus may be a 14-bit bus to accommodate the command/address signals (CA<13:0>). The command/address signals are clocked to the command interface **14** using the clock signals (Clk_t and Clk_c). The command interface may include a command address input circuit **20**, which is configured to receive and transmit the commands to provide access to the memory banks **12**, through the command decoder **32**, for instance. In addition, the command interface **14** may receive a chip select signal (CS_n). The CS_n signal enables the memory device **10** to process commands on the incoming CA<13:0> bus. Access to specific banks **12** within the memory device **10** is encoded on the CA<13:0> bus with the commands.

[0029] In addition, the command interface **14** may be configured to receive a number of other command signals. For instance, a command/address on die termination (CA_ODT) signal may be provided to facilitate proper impedance matching within the memory device **10**. A reset command (RESET_n) may be used to reset the command interface **14**, status registers, state machines and the like, during power-up for instance. The command interface **14** may also receive a command/address invert (CAI) signal which may be provided to invert the state of command/address signals CA<13:0> on the command/address bus, for instance, depending on the command/address routing for the particular memory device **10**. A mirror (MIR) signal may also be provided to facilitate a mirror function. The MIR signal may be used to multiplex signals so that they can be swapped for enabling certain routing of signals to the memory device **10**, based on the configuration of multiple memory devices in a particular application. Various signals to facilitate testing of the memory device **10**, such as the test enable (TEN) signal, may be provided, as well. For instance, the TEN signal may be used to place the memory device **10** into a test mode for connectivity testing.

[0030] The command interface **14** may also be used to provide an alert signal (ALERT_n) to the system processor or controller for certain errors that may be detected. For instance, an alert signal (ALERT_n) may be transmitted from the memory device **10** if a cyclic redundancy check (CRC) error is detected. Other alert signals may also be generated. Further, the bus and pin for transmitting the alert signal (ALERT_n) from the memory device **10** may be used as an input pin during certain operations, such as the connectivity test mode executed using the TEN signal, as described above.

[0031] Data may be sent to and from the memory device **10**, utilizing the command and clocking

signals discussed above, by transmitting and receiving data signals **44** through the IO interface **16**. More specifically, the data may be sent to or retrieved from the memory banks **12** over the datapath **46**, which includes a plurality of bi-directional data buses. Data IO signals, generally referred to as DQ signals, are generally transmitted and received in one or more bi-directional data busses. For certain memory devices, such as a DDR5 SDRAM memory device, the IO signals may be divided into upper and lower bytes. For instance, for a x16 memory device, the IO signals may be divided into upper and lower IO signals (e.g., DQ<15:8> and DQ<7:0>) corresponding to upper and lower bytes of the data signals, for instance.

[0032] To allow for higher data rates within the memory device **10**, certain memory devices, such as DDR memory devices may utilize data strobe signals, generally referred to as DQS signals. The DQS signals are driven by the external processor or controller sending the data (e.g., for a write command) or by the memory device **10** (e.g., for a read command). For read commands, the DQS signals are effectively additional data output (DQ) signals with a predetermined pattern. For write commands, the DQS signals are used as clock signals to capture the corresponding input data. As with the clock signals (Clk_t and Clk_c), the DQS signals may be provided as a differential pair of data strobe signals (DQS_t and DQS_c) to provide differential pair signaling during reads and writes. For certain memory devices, such as a DDR5 SDRAM memory device, the differential pairs of DQS signals may be divided into upper and lower data strobe signals (e.g., UDQS_t and UDQS_c; LDQS_t and LDQS_c) corresponding to upper and lower bytes of data sent to and from the memory device **10**, for instance.

[0033] An impedance (ZQ) calibration signal may also be provided to the memory device **10** through the IO interface **16**. The ZQ calibration signal may be provided to a reference pin and used to tune output drivers and ODT values by adjusting pull-up and pull-down resistors of the memory device **10** across changes in process, voltage, and temperature (PVT) values. Because PVT characteristics may impact the ZQ resistor values, the ZQ calibration signal may be provided to the ZQ reference pin to be used to adjust the resistance to calibrate the input impedance to known values. As will be appreciated, a precision resistor is generally coupled between the ZQ pin on the memory device **10** and GND/VSS external to the memory device **10**. This resistor acts as a reference for adjusting internal ODT and drive strength of the IO pins.

[0034] In addition, a loopback data signal (LBDQ) and loopback strobe signal (LBDQS) may be provided to the memory device **10** through the IO interface **16**. The loopback data signal and the loopback strobe signal may be used during a test or debugging phase to set the memory device **10** into a mode wherein signals are looped back through the memory device **10** through the same pin. For instance, the loopback signal may be used to set the memory device **10** to test the data output (DQ) of the memory device **10**. Loopback may include both LBDQ and LBDQS or possibly just a loopback data pin. This is generally intended to be used to monitor the data captured by the memory device **10** at the IO interface **16**. LBDQ may be indicative of a target memory device, such as memory device **10**, data operation and, thus, may be analyzed to monitor (e.g., debug and/or perform diagnostics on) data operation of the target memory device. Additionally, LBDQS may be indicative of a target memory device, such as memory device **10**, strobe operation (e.g., clocking of data operation) and, thus, may be analyzed to monitor (e.g., debug and/or perform diagnostics on) strobe operation of the target memory device.

[0035] As will be appreciated, various other components such as power supply circuits (for receiving external VDD and VSS signals), mode registers (to define various modes of programmable operations and configurations), read/write amplifiers (to amplify signals during read/write operations), temperature sensors (for sensing temperatures of the memory device **10**), etc., may also be incorporated into the memory device **10**. Accordingly, it should be understood that the block diagram of FIG. **1** is only provided to highlight certain functional features of the memory device **10** to aid in the subsequent detailed description.

[0036] FIG. **2** is a top view **60** of lines of ferroelectric memory cells of FIG. **1** and a side view **74**

of the lines in the memory cells of FIG. 1 with additional connections. As illustrated, a ferroelectric memory device (e.g., memory device **10**) may utilize a plate line (PL) **62** that provides a plate voltage to multiple cells in a row/column. Each PL is flanked by two digit lines that are utilized for cells connected to the respective PL. For instance, the PL **62** is flanked by DL **64** and DLB **66** at each cell along the PL **62**. For instance, PL **68** may be flanked by DL **70** and DLB **72**. Other plate lines may be used for other cells in other rows/columns.

[0037] In the side view **74**, the PLs **62** and **68** utilize a wordline (WL) **76** to control access to respective cells. As illustrated, the PL **62** is sandwiched between ferroelectric layers **78** and **80**. The ferroelectric layers **78** and **80** are adjacent to opposite sides of the PL **62**. For instance, the ferroelectric layers **78** and **80** may be formed from ferroelectric oxides, lead zirconium titanate, and/or other materials that have ferroelectric properties. Opposite the sides of the ferroelectric layers **78** and **80** coupled to and adjacent to the PL **62**, the ferroelectric layers **78** and **80** are coupled to conductors **82** and **84** that may be extensions of source or drain terminals of respective selectors **86** and **88**. The selectors **86** and **88** may be transistors. Instead, as discussed below, a write back circuit that works with higher voltages may be used to write back to the DL **64** and/or the DLB **66**. The ferroelectric layer **78** connected to the PL **62** and the conductor **82** form a first cell while the ferroelectric layer **80** connected to the PL **62** and the conductor **84** form a second cell. The selectors **86** and **88** select the respective cells using the WL **76**.

[0038] Similarly, the PL **68** is sandwiched between ferroelectric layers **90** and **92** with the ferroelectric layers **90** and **92** adjacent to opposite sides of the PL **68**. Like the ferroelectric layers **78** and **80**, the ferroelectric layers **90** and **92** may be formed from ferroelectric oxides, lead zirconium titanate, and/or other materials that have ferroelectric properties. Opposite the sides of the ferroelectric layers **90** and **92** coupled to and adjacent to the PL **68**, the ferroelectric layers **90** and **92** are coupled to conductors **94** and **96** that may be extensions of source or drain terminals of respective selectors **98** and **100**. The selectors **98** and **100** may be of the same type, materials, and/or characteristics as the selectors **86** and **88**. The ferroelectric layer **90** connected to the PL **68** and the conductor **94** form a first cell while the ferroelectric layer **92** connected to the PL **68** and the conductor **96** form a second cell. The selectors **98** and **100** select the respective cells using the WL **76**.

[0039] Although the foregoing discusses using a PL for the cells, in some embodiments, at least two of the PLs (e.g., PL **62** and **68**) may be merged into a single plate **102** shared between at least 4 digit lines (e.g., DL **64**, DLB **66**, DL **70**, and DLB **72**).

[0040] FIG. 3 is a circuit diagram of a connection **120** of the PL **62** through ferroelectric layers **78** and **80** via selectors **86** and **88** to the DL **64** and the DLB **66**, respectively. The DL **64** and the DLB **66** are received at a sense amplifier **122** as a differential signal. The sense amplifier **122** may be one of the sense amplifiers **13** of FIG. 1. The sense amplifier **122** utilizes this differential signal to output a value **124**. In some embodiments, the sense amplifier **122** may also send a complementary value along with the value **124**.

[0041] Memory devices (e.g., FeRAM) may be implemented using 1T1C or 2T2C configurations. 2T2C implementations provide a higher probability of a correct interpretation and provides faster speeds. However, the 2T2C implementation consumes more resources regarding financial costs, power costs, and area costs. The decision to use 1T1C and 2T2C may vary based on particular use cases.

[0042] FIG. 4 is a block diagram of the sense amplifier **13** or **122**. The sense amplifier **13** includes a current source **216** that provides a bias current and/or a bias voltage (e.g., up to 1.2V). The sense amplifier **13** also includes switch **217** that uses a control₁ signal **218** to selectively couple the current source **216** to terminals (e.g., drain or source terminals) of PMOS transistors **219** and **220**. In some embodiments, the terminals of the PMOS transistors **219** and **220** may be coupled to one or more current sources **216** using separate switches each controlled by the control₁ signal **218**.

[0043] The gate terminal of the PMOS transistor **219** is coupled to a gate node **211** of the sense

amplifier **13**, and the gate terminal of the PMOS transistor **220** is coupled to a gut node **213** of the sense amplifier **13**. The other terminal (e.g., drain or source) of the PMOS transistor **219** is coupled to a switch **221** and a switch **224**. The other terminal (e.g., drain or source) of the PMOS transistor **220** is coupled to a switch **222** and a switch **225**. The switches **221** and **222** are controlled by a control2 signal **223** that selectively couples the respective other terminals of the PMOS transistors **219** and **220** to their respective gate terminals. As discussed below, this selectively coupling makes a negative feedback amplifier/inverting amplifier/inverter equivalent to a unit gain amplifier.

[0044] The switches **224** and **225** are controlled by a control3 signal **226**. Based on the control3 signal **226**, the switch **224** selectively couples the other terminal of the PMOS transistor **219** to the gut node **213**. Similarly, based on the control3 signal **226**, the switch **225** selectively couples the other terminal of the PMOS transistor **220** to the gut node **211**.

[0045] The sense amplifier **13** also includes switches **227** and **228** that are controlled using a control4 signal **229**. Based on the control4 signal **229**, the switch **227** selectively couples the gut node **213** to the DL **64**. Similarly, based on the control4 signal **229**, the switch **228** selectively couples the gut node **211** to the DLB **66**. In other words, the control4 signal **229** acts as an isolation signal that may be used to selectively decouple the sense amplifier **13** from the DL **64** and the DLB **66**.

[0046] The sense amplifier **13** further includes a switch **230** that utilizes a control5 signal **232** to selectively couple and decouple a terminal (e.g., drain) of NMOS transistors **233** and **234** to V.sub.neg **235**. In FeRAM devices, the operating voltage cannot just be a middle point (e.g., 0.5V) between the upper and lower voltages (VCC (e.g., 1V) at the current source **216** and VSS (e.g., 0V)). This middle point strategy between two positive voltages is not readily available for FeRAM devices because the ferroelectric layers **90** and **92** must rest near 0V. Accordingly, Vneg **235** may be a voltage level below 0V or may be any voltage lower than a voltage between the switch **217** and the PMOS transistors **219** and/or **220**. A terminal of the NMOS transistor **233** is coupled to the terminal of the PMOS transistor **219** and to the switches **221** and **224**, and a terminal of the NMOS transistor **234** is coupled to a terminal of the PMOS transistor **220** and to switches **222** and **225**. The gate of the NMOS transistor **233** is coupled to the gate of the PMOS transistor **219** and to the gut node **213**, and the gate of the NMOS transistor **234** is coupled to the gate of the PMOS transistor **220** and to the gut node **211**.

[0047] As discussed below, the sense amplifier **13** may apply a deflection voltage **236** that enables the sense amplifier **13** to help track the voltage difference in the memory cells by increasing the voltage difference between the DL **64** and the PL **62**. The deflection voltage **236** may utilize a counter to inject current to shift the DL **64** and the DLB **66** lower to aid in extracting the voltage from the PL **62** to the DL **64**. The deflection voltage **236** may be applied to the DL **64** and the DLB **66** via capacitors **237** and **238**. The capacitors may be MOSFET devices that are configured as capacitors.

[0048] Additionally, the sense amplifier **13** may include some capacitance on the lines coupled to the gut nodes **213** and **211** that are illustrated as capacitors **240** and **242**. Although these capacitors **240** and **242** are shown as distinct capacitive devices, the capacitance may be parasitic capacitance of the lines connecting various components of the sense amplifier **13**.

[0049] Furthermore, the sense amplifier **13** includes a transistor **244** (e.g., an NMOS or PMOS transistor) that selectively couples the DL **64** to ground (e.g., VSS) based on a control6 signal **246** to drive (e.g., discharge charge from) the DL **64** toward the connected voltage via the transistor **244**. In some embodiments, the voltage selectively coupled to the DL **64** via the transistor **244** may be different than VSS/ground and may be any voltage to which the DL **64** may be driven. Likewise, the sense amplifier **13** includes a transistor **248** (e.g., an NMOS or PMOS transistor) that selectively couples the DLB **66** to ground (e.g., VSS) based on the control6 signal **246** to drive (e.g., discharge charge from) the DLB **66** toward the connected voltage via the transistor **248**. In some embodiments, the voltage selectively coupled to the DLB **66** via the transistor **248** may be

different than VSS/ground and may be any voltage to which the DLB 66 may be driven. In some embodiments, the voltages selectively coupled to the DL 64 via the transistor 244 and to the DLB 66 via the transistor 248 may be the same as each other or may be different from each other.

[0050] As is discussed in more detail below, the PMOS transistor 219 and the NMOS transistor 233 are coupled together as an inverter 265 (e.g., complementary metal oxide semiconductor (CMOS) inverter) with the voltage of the gut node 211 as an input to the inverter 265 and with an output to the switch 224. Similarly, the PMOS transistor 220 and the NMOS transistor 234 are coupled together as an inverter 266 (e.g., CMOS inverter) with the voltage of the gut node 213 as an input to the inverter 266 and with an output to the switch 225.

[0051] Although the switches 217, 221, 222, 224, 225, 227, 228, and 230 are illustrated as generic switches, they may be implemented with any suitable switch embodiment. For instance, for the purposes of the following discussion, the switch 217 is implemented using a PMOS transistor while the switches 221, 222, 224, 225, 227, 228, and 230 are implemented using NMOS transistors. However, other implementations for these switches may be used in other embodiments with the respective control signals changed accordingly.

[0052] FIG. 5 shows a graph 250 and a timing diagram 252 for utilizing the sense amplifier 13 across phases 254, 256, 258, and 260. The phase 254 is a VTH compensation phase where the respective characteristics or “VTHs” of the inverters 265 and 266 are compensated for by putting sufficient charge on the DL 64 for the inverter 266 and on the DLB 66 for the inverter 265 to cause the inverters 265 and 266 to respond similarly by enhancing/optimizing the gains for the inverters 265 and 266 on their respective input-output curves. The phase 254 includes disconnecting the DL 64 and the DLB 66 from voltages (e.g., VSS) using the control6 signal 246 via the transistors 244 and 248 to enable charging of the DL 64 and the DLB 66 corresponding to the characteristics of the inverters 265 and 266. The phase 254 also includes turning on the switch 217 via the control1 signal 218 transitioning low to turn on the switch 217 (PMOS transistor). The control5 signal 232 is also asserted (high for NMOS transistor) so that, along with assertion of the control1 signal 218, the bias current is enabled to flow. The control2 signal 223 is also asserted (e.g., high) during the phase 254 causing switches 221 and 222 to activate. The activation of the switches 221 and 222 (NMOS transistors) causes the inputs and outputs of the inverters 265 and 266 to be tied together causing the inverters 265 and 266 to function as a unity gain amplifier that accelerates the settling of the voltages of the DL 64 and the DLB 66. The control3 signal 226 is low causing the switches 224 and 225 (NMOS transistors) to be off thereby decoupling the inverter 265 from the gut node 213 and decoupling the inverter 266 from the gut node 211. The control4 signal 229 is also high causing the switches 227 and 228 (NMOS transistors) to be on thereby respectively connecting the gut nodes 213 and 211 to the DL 64 and the DLB 66. This enables charges to be stored on the DL 64 corresponding to and proportional to the characteristics (or VTH) of the inverter 266 and on the DLB 66 corresponding to the characteristics (or VTH) of the inverter 265. The stored “VTH” may include the bias from the current source 216.

[0053] Further, during phase 254, the WL 76 turns on. The WL 76 is represented in the graph 250 as a WL_near 76A for WLs near the WL driver and as WL_far 76B for WLs far from the WL driver of the memory device 10. The WL 76 may be turned on during the phase 254 to improve tRCD (i.e., time to open a row and access a column of memory cells). This early turn on anticipates the RC delays to the voltage evolution of the WL_far 76B. The PL 62 also remains off. Additionally, during phase 254, the control5 signal 232 is high to turn off the switch 230.

[0054] After the phase 254 and at the beginning of the phase 256, the control1 signal 218 transitions high, and the control5 signal 232 transitions low. The control1 signal 218 and the control5 signal 232 transitioning causes the bias current to be cut off and the gut nodes 211 and 213 to be in a high impedance state.

[0055] After a delay 261 enabling the transition of the control1 signal 218 and the control5 signal 232 to complete, the control2 signal 223 is de-asserted (e.g., low for NMOS) causing the inverters

265 and **266** to no longer be configured to function as unity gain amplifiers. Also, during phase **256**, the PL **62** transitions high, and the deflection voltage **236** is also issued to the sense amplifier **13** via the capacitors **237** and **238**. In some embodiments, as illustrated, the sense amplifier **13** may apply the deflection voltage **236** during the phase **254** in addition or in alternative to applying the deflection voltage **236** during the phase **256**.

[0056] During the phase **256**, the connected memory cells are biased, and their charges are dumped on the DL **64** and the DLB **66** and a voltage is developed on the DL **64**, the DLB **66**, and the gut nodes **211** and **213**, as shown in the graph **250**.

[0057] Also, as illustrated in the timing diagram **252**, the deflection voltage **236** is reduced during the phase **256** after time is given for the voltage difference between the gut nodes **211** and **213** to develop. This reduction of the deflection voltage **236** enables the overall voltage to be lowered for the gut nodes **211** and **213**, the DL **64**, and the DLB **66** without negatively impacting the voltage difference and fidelity of the output of the sense amplifier **13**.

[0058] During the phase **258**, the signal in the sense amplifier **13** is amplified by the NMOS transistors. Specifically, the control4 signal **229** is used to disconnect the DL **64** from the gut node **213** and to disconnect the DLB **66** from the gut node **211**. After a delay **262** enabling the control4 signal **229** to isolate the gut nodes **211** and **213** from the DL **64** and DLB **66**, the control6 signal **246** is used to set the DL **64** and the DLB **66** to some voltage (e.g., VSS) via transistors **244** and **248** without impacting the voltages of the gut nodes **211** and **213**. Likewise, after the delay **262**, the control5 signal **232** toggles the switch **230** to enable amplification on the gut node **211** and **213** using the NMOS devices.

[0059] At the beginning of phase **260**, the control1 signal **218** is asserted to enable PMOS amplification while NMOS amplification continues from the phase **258** into the phase **260**. In other words, the connection of the NMOS transistors **233** and **234** with the PMOS transistors **219** and **220** completes the latch connection. In such a configuration, the inverters **265** and **266** (with their respective PMOS transistors **219** and **220**) amplify (e.g., negative amplification) the signal on the gut nodes **211** and **213** due to VTH compensation enhancing/optimizing the responses of the inverters **265** and **266** specific to their own characteristics. Due to this VTH compensation of the inverters **265** and **266**, the difference in the values stored from the cells may be greatly increased. Furthermore, any error that may be introduced in sensing may be lower than the signals on the gut nodes **211** and **213** after amplification and/or than the voltage differences between the gut nodes **211** and **213** is increased.

[0060] FIG. **6** is a simplified diagram of an inverter **265A** that is a visualization of the inverter **265**. The inverter **265** operates using the PMOS transistor **220** and the NMOS transistor **234** coupled as a CMOS inverter between the current source **216** for the bias current. The inverter **265** includes a switch **222** couples the input (e.g., gut node **211**) of the inverter **265** with an output **270** of the inverter **265**. The output **270** of the inverter **265** may be coupled to the switch **225** and/or the gut node **213** depending on whether the switch **225** is open or closed.

[0061] FIG. **6** also includes a simplified diagram of an inverter **265B** that is another visualization of the inverter **265** during the VTC operation in the phase **254**. The inverter **265B** shows the inverter **265** as an inverter **271** (e.g., inverting amplifier) that has a feedback loop from the output **270** to the input (e.g., gut node **211**) when the switch **222** is closed. FIG. **6** further depicts a graph **272** depicting a response curve plotting a voltage input **274** versus a voltage output **275**. The graph **272** includes an input response curve **276** that plots the input voltages to the inverter **265B** based on the input voltages along the x-axis of the graph and an output response curve for a unity gain amplifier. The graph **272** further includes an output response curve **277** that indicates a voltage of the output **270** based on the input voltages. This output response curve **277** may vary from one implementation of the inverter **265** to another embodiment of the inverter **265** due to specific characteristics while another implementation of the inverter **265** corresponding to output response curve **278** may have different characteristics even if the same design. For instance, the thickness of

the transistors may vary even by a very small degree/percentage, the transistors of the different implementations may have aged differently, and/or any other properties between the characteristics may vary. The different output response curves **277** and **278** may have different responses at different voltages as the gain of the inverters **265B** may be $-A$ that is the slope or ratio of the change (e.g., negative) in output voltage divided by the change (e.g., positive) in input voltage. Due to these differences in slope at a specific input voltage (e.g., at line **279**), the implementation corresponding to the output response curve **277** may have a greater negative gain than the output response curve **278** for the same input voltage.

[0062] One way to minimize these variances between different implementations of the inverters **265** or **266**, the VTC compensation in the phase **254** may be used to place both inverters **265** and **266** at an improved or ideal location near the centers of the output response curves **277** and **278** to enhance or even optimize amplification. This may be done by shorting the inputs and outputs of the inverters **265** and **266** during the phase **254** by configuring the inverters **265** and **266** as unity gain amplifiers by switching switches **221** and **222**. Once the feedback is removed by opening the switches **221** and **222**, the voltage remains the same at the output at this point of equilibrium where the corresponding output response curve **277** or **278** matches the input response curve **276**. Subsequently, when the input voltage changes in the subsequent phases **256-260**, the corresponding output value moves more due to placing the output voltage at the intersection of the input response curve **276** and the corresponding output response curve **277** or **278**. At this point, the gain is at or near (e.g., within 5%, 10%, 25% or another percentage of) the maximum amplification value. This greater amplification causes a greater difference between the differential signal from the memory cells before being dumped back on the DL **64** and the DLB **66**. This greater difference causes the data stored in the memory cells to be properly sensed in the sense amplifier **13**.

[0063] FIG. **7** is a schematic diagram of an embodiment of the sense amplifiers **13**, **122**. The embodiment in FIG. **7** is similar to the embodiment in FIG. **4** except that the embodiment of FIG. **7** uses a ground connection **280** instead of Vneg **235** at a negative voltage. However, to enable the required rest near 0V for FeRAM, the sense amplifier of FIG. **7** includes decoupling capacitors **282** and **284** that are used to bring the voltages of the DL **64** and the DLB **66** to at or around 0V. Thus, the voltage across the decoupling capacitors **282** and **284** is the difference between at or around 0V and an offset at the gut nodes **211** and **213** at least partially attributable to an offset from the current source **216**.

[0064] FIG. **8** shows a graph **290** and the timing diagram **252** for utilizing the sense amplifier **13** across phases **254**, **256**, **258**, and **260**. The graph **290** is similar to the graph **250** except that the voltages of the gut nodes **211** and **213** are offset from the voltages of the DL **64** and the DLB **66** by the decoupling capacitors **282** and **284**. For instance, the voltage of the gut node **211** may be greater than the voltage of the DL **64** by an offset voltage **292** across the decoupling capacitor **282**. Likewise, the voltage of the gut node **213** may be greater than the voltage of the DLB **66** by an offset voltage **294** across the decoupling capacitor **284**. Thus, the gut voltages may operate at a mid-point voltage between an upper voltage and 0V while the voltage on the DL **64** and DLB **66** rests at or near 0V. For instance, before phase **254** and at the end and after the end of the phase **260**, the DL **64** and the DLB **66** may be at or around 0V. The timing diagram **252** for FIGS. **5** and **8** are the same.

[0065] While the present disclosure may be susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, it should be understood that the present disclosure is not intended to be limited to the particular forms disclosed. Rather, the present disclosure is intended to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the following appended claims.

[0066] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field

and, as such, are not abstract, intangible, or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112 (f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112 (f).

Claims

1. A memory device, comprising: a plate line; a pair of ferroelectric layers implementing a pair of memory cells and coupled to opposite sides of the plate line; a pair of digit lines each coupled to a respective ferroelectric layer of the pair of ferroelectric layers; and a sense amplifier coupled to the pair of digit lines and configured to sense and amplify voltages received at the pair of digit lines from the respective memory cells, wherein the sense amplifier comprises: a pair of inverters configured to selectively couple to the pair of digit lines; and a pair of switches each configured to cause a respective inverter of the pair of inverters to function as a unity gain amplifier during a compensation phase of the memory device.
2. The memory device of claim 1, wherein the pair of inverters comprises: a first complementary metal-oxide semiconductor inverter; and a second complementary metal-oxide semiconductor inverter.
3. The memory device of claim 1, wherein each of the pair of switches is configured to connect an input of a respective inverter of the pair of inverters to a corresponding output of the respective inverter of the pair of inverters as a feedback connection.
4. The memory device of claim 1, wherein the memory device comprises: a current source; and a current switch that is configured to couple source terminals of p-channel transistors of the pair of inverters to the current source during threshold compensation using the compensation phase.
5. The memory device of claim 1, wherein the compensation phase is configured to cause the pair of inverters to amplify input values similarly during one or more amplification phases.
6. The memory device of claim 1, wherein the compensation phase is configured to settle input and output voltages of the pair of inverters to enhance gains of the pair of inverters.
7. The memory device of claim 6, wherein the gains are negative gains.
8. The memory device of claim 6, wherein the gains of the pair of the inverters are optimized at respective locations on respective characteristic-dependent output response curves corresponding to a greatest gain of the respective characteristic-dependent output response curves.
9. The memory device of claim 1, wherein the memory device comprises an isolation pair of switches to couple gate nodes of the sense amplifier to respective digit lines of the pair of digit lines during the compensation phase.
10. The memory device of claim 1, wherein the compensation phase comprises charging respective digit lines of the pair of digit lines with values specific to characteristics of respective inverters of the pair of inverters.
11. A memory device, comprising a plurality of ferroelectric layer-based memory cells to store data; a plurality of first digit lines each coupled to a respective ferroelectric layer-based memory cell of a first set of the plurality of ferroelectric layer-based memory cells; a plurality of second digit lines each coupled to a respective ferroelectric layer-based memory cell of a second set of the plurality of ferroelectric layer-based memory cells; and a plurality of sense amplifiers each respectively coupled to a first digit line from the plurality of first digit lines and to a second digit line from the plurality of second digit lines, wherein each of the plurality of sense amplifiers comprises a pair of inverters configured to selectively couple to the first and second digit lines, and each inverter of the pair of inverters comprises a switch configured to cause the respective inverter to function as a unity gain amplifier during a compensation phase of the memory device.

- 12.** The memory device of claim 11, wherein each of the sense amplifiers of the plurality of sense amplifiers is configured to sense and amplify a voltage difference between voltages on the respective first and second digit lines.
 - 13.** The memory device of claim 11, wherein each of the first set of the plurality of ferroelectric layer-based memory cells is configured to store data, and each of the second set of the plurality of ferroelectric layer-based memory cells is configured to store complementary data.
 - 14.** The memory device of claim 13, wherein the first digit lines are configured to carry the data.
 - 15.** The memory device of claim 14, wherein the second digit lines are configured to carry a complement of the data.
 - 16.** The memory device of claim 11, comprising a current source that selectively supplies a current to the pair of inverters.
 - 17.** The memory device of claim 11, wherein each of the switches is configured to couple together an input and an output of the respective inverter during the compensation phase before one or more amplification phases of the respective sense amplifier.
 - 18.** A method, comprising: toggling a first switch in a sense amplifier during a compensation phase to couple a first input of a first inverter to a first output of the first inverter to cause the first inverter of the sense amplifier to be configured as a unity gain amplifier during the compensation phase of the sense amplifier to cause amplification in the first inverter to be enhanced in subsequent amplification phases of the sense amplifier by storing a first charge on a first digit line of the sense amplifier; toggling a second switch in the sense amplifier during the compensation phase to couple a second input of a second inverter to a second output of the second inverter to cause the second inverter of the sense amplifier to be configured as a unity gain amplifier during the compensation phase of the sense amplifier to cause amplification in the second inverter to be enhanced in the subsequent amplification phases of the sense amplifier by storing a second charge on a second digit line of the sense amplifier; in an amplification phase, disconnecting the first input from the first output; in the amplification phase, disconnecting the second input from the second output; and amplifying a signal difference from memory cells corresponding to the sense amplifier by using the enhanced amplification in the first and second inverters.
 - 19.** The method of claim 18, comprising applying a deflection voltage to reduce voltages in the first and second digit lines during a voltage level development between the compensation phase and the subsequent amplification phases.
 - 20.** The method of claim 19, comprising applying the deflection voltage via capacitors coupled to the first and second digit lines.
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